

AI/ML Internship – Day 1 Submission

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Role: AI/ML Intern

Project Folder: Raritone

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1. Introduction

Artificial Intelligence and Computer Vision enable machines to interpret visual data and understand human actions. As part of Day 1 of the internship, the objective was to set up the development environment, study key AI concepts, and implement a basic real-time pose detection system.

This report includes research on pose detection, body measurement AI, and virtual try-on technology, along with a practical implementation using Python.

2. Pose Detection

Overview

Pose detection is a computer vision technique that identifies key points of the human body from images or video. These key points include joints such as shoulders, elbows, hips, and knees. By connecting these points, a skeletal structure is formed, allowing systems to understand posture and movement.

Working Principle

- Capture image or video using a camera
- Detect the human body in the frame
- Identify body landmarks (joints)
- Connect these points to form a skeleton
- Analyze movement and posture

Pose detection is commonly implemented using tools like MediaPipe along with OpenCV and Python.

Applications

- Fitness and exercise tracking
- Posture correction and yoga monitoring
- Gaming and animation
- Healthcare and physiotherapy
- Virtual dressing systems

3. AI in Fashion Shopping

Overview

Artificial Intelligence has significantly improved the fashion industry by making shopping more efficient and personalized. It enhances both user experience and business operations.

Key Applications

Virtual Try-On

Allows users to digitally try clothes using camera input or images.

Body Measurement AI

Estimates body size and suggests appropriate clothing sizes, reducing size mismatch.

Personalized Recommendations

Recommends outfits based on user preferences, browsing history, and trends.

Visual Search

Enables users to search for products using images instead of text.

Trend Analysis

Helps companies analyze data and predict fashion trends for better inventory management.

Benefits

- Improved customer experience
- Accurate size prediction
- Faster and easier product discovery
- Reduced product returns

4. Implementation

A real-time pose detection system was implemented using OpenCV and MediaPipe.

Steps Performed

- Accessed webcam using Python
- Captured live video frames
- Converted frames for processing
- Detected body landmarks using the pose model
- Displayed skeletal connections in real time

5. Code Implementation

```
import cv2
import mediapipe as mp

mp_pose = mp.solutions.pose
mp_draw = mp.solutions.drawing_utils

pose = mp_pose.Pose(
    static_image_mode=False,
    model_complexity=2,
    min_detection_confidence=0.3,
    min_tracking_confidence=0.3
)

cap = cv2.VideoCapture(0)

while True:
    success, frame = cap.read()
    if not success:
        break

    frame = cv2.flip(frame, 1)

    rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
    results = pose.process(rgb)
```

```

if results.pose_landmarks:
    mp_draw.draw_landmarks(
        frame,
        results.pose_landmarks,
        mp_pose.POSE_CONNECTIONS
    )

cv2.imshow("Pose Detection", frame)

if cv2.waitKey(1) & 0xFF == ord('q'):
    break

cap.release()
cv2.destroyAllWindows()

```

Tools and Technologies Used

The following tools and libraries were installed and used during this task:

- **Python**
Used as the primary programming language for implementing AI/ML concepts.
- **Visual Studio Code**
Used as the development environment for writing and executing code.
- **TensorFlow**
Installed as a deep learning framework (used internally by many AI models).
- **OpenCV (cv2)**
Used for capturing webcam input and processing images/video frames.
- **MediaPipe**
Used for detecting human body landmarks and performing pose estimation.

